

CITYFIX BUMI SHOLAWAT

Tahap 3 - Testing & Improvement

Panduan testing, UAT, security review, bug fixing, dan improvement

Dokumen ini melanjutkan Tahap 2. Fokusnya adalah menguji fungsi, keamanan role, validasi input, upload foto, workflow, akses data, usability mobile, serta menyiapkan UAT dan regression test sebelum aplikasi masuk production.

1. Target Akhir Tahap 3

Tahap 3 bertujuan memastikan aplikasi CityFix yang sudah selesai dikembangkan pada Tahap 2 benar secara fungsi, aman berdasarkan role, stabil, mudah digunakan, dan siap diuji oleh pengguna nyata.

```
Reporter -> Login -> Buat Laporan -> Foto tersimpan
-> Verifier verifikasi + assign teknisi
-> Teknisi update status -> upload bukti
-> Completed -> Reporter melihat hasil
-> Dashboard dan histori ikut berubah
```

Selain alur normal, sistem juga harus menolak akses tidak sah dan input yang tidak valid.

Bagian	Fokus
3.1	Environment & Database Test
3.2	Functional Testing
3.3	Role & Security Testing
3.4	Validation & Upload Testing
3.5	User Acceptance Test (UAT)
3.6	Improvement & Bug Fix

2. Persiapan Database Testing

Jangan jalankan automated test pada database development utama. Buat database terpisah supaya data development tidak terhapus atau tercampur.

```
CREATE DATABASE cityfix_testing;
```

2.1 Konfigurasi phpunit.xml

```
<env name="APP_ENV" value="testing"/>
<env name="DB_CONNECTION" value="mysql"/>
<env name="DB_DATABASE" value="cityfix_testing"/>
<env name="DB_USERNAME" value="root"/>
<env name="DB_PASSWORD" value=""/>
```

Pastikan nama database, username, dan password sesuai dengan MySQL Anda.

2.2 Menjalankan test dasar

```
php artisan test
```

Jika test bawaan Laravel PASS, environment testing sudah berjalan.

3. Factory untuk Data Test

Factory membuat data dummy secara otomatis sehingga setiap test dapat berdiri sendiri.

3.1 UserFactory

```
public function definition(): array
{
    return [
        'name' => fake()->name(),
        'email' => fake()->unique()->safeEmail(),
        'email_verified_at' => now(),
        'password' => bcrypt('password'),
        'role' => 'reporter',
        'remember_token' => null,
    ];
}

public function admin() { return $this->state(fn () => ['role'=>'admin']); }
public function verifier() { return $this->state(fn () => ['role'=>'verifier']); }
public function technician() { return $this->state(fn () => ['role'=>'technician']); }
public function reporter() { return $this->state(fn () => ['role'=>'reporter']); }
```

3.2 AreaFactory

```
php artisan make:factory AreaFactory --model=Area

return [
    'name' => fake()->unique()->words(2, true),
    'code' => fake()->unique()->bothify('AR-###'),
    'is_active' => true,
];
```

3.3 CategoryFactory

```
php artisan make:factory CategoryFactory --model=Category

return [
    'name' => fake()->unique()->word(),
    'icon' => null,
    'is_active' => true,
];
```

3.4 ReportFactory

```
php artisan make:factory ReportFactory --model=Report

return [
    'report_number' => 'CF-'.fake()->unique()->numerify('#####'),
    'user_id' => User::factory(),
    'area_id' => Area::factory(),
    'category_id' => Category::factory(),
    'location_detail' => fake()->sentence(3),
    'description' => fake()->sentence(),
    'urgency' => fake()->randomElement(['low', 'medium', 'high', 'emergency']),
    'status' => 'reported',
    'photo' => null,
    'assigned_to' => null,
    'reported_at' => now(),
];
```

4. Authentication Test

```
php artisan make:test AuthenticationTest
```

4.1 Login page

```
public function test_login_page_can_be_opened(): void
{
    $response = $this->get('/login');
    $response->assertStatus(200);
}
```

4.2 Login valid

```
public function test_user_can_login(): void
{
    $user = User::factory()->create(['password'=>bcrypt('password123')]);
    $response = $this->post('/login', [
        'email'=>$user->email,
        'password'=>'password123',
    ]);
    $response->assertRedirect('/dashboard');
    $this->assertAuthenticatedAs($user);
}
```

4.3 Password salah

```
$response = $this->post('/login', [
    'email'=>$user->email,
    'password'=>'wrong-password',
]);
$response->assertSessionHasErrors('email');
$this->assertGuest();
```

4.4 Protected page

```
public function test_guest_cannot_open_dashboard(): void
{
    $this->get('/dashboard')->assertRedirect('/login');
}
```

```
public function test_authenticated_user_can_open_dashboard(): void
{
    $user = User::factory()->create();
    $this->actingAs($user)->get('/dashboard')->assertStatus(200);
}
```

5. Functional Test - Create Report

```
php artisan make:test ReportTest
```

5.1 Halaman create report

```
$user = User::factory()->reporter()->create();
$this->actingAs($user)->get('/reports/create')->assertStatus(200);
```

5.2 Upload foto dan simpan laporan

```
Storage::fake('public');
$user = User::factory()->reporter()->create();
$area = Area::factory()->create();
$category = Category::factory()->create();
$photo = UploadedFile::fake()->image('keran-bocor.jpg', 800, 600);

$response = $this->actingAs($user)->post('/reports', [
    'area_id'=>$area->id,
    'category_id'=>$category->id,
    'location_detail'=>'Toilet lantai 2',
    'description'=>'Keran terus mengeluarkan air.',
    'urgency'=>'medium',
    'photo'=>$photo,
]);

$response->assertSessionHasNoErrors();
$this->assertDatabaseHas('reports', [
    'user_id'=>$user->id,
    'area_id'=>$area->id,
    'status'=>'reported',
]);
```

5.3 File bukan gambar harus ditolak

```
$file = UploadedFile::fake()->create('dokumen.pdf', 500, 'application/pdf');
// submit seperti biasa
$response->assertSessionHasErrors('photo');
```

5.4 File terlalu besar

```
$photo = UploadedFile::fake()->image('besar.jpg')->size(6000);
$response->assertSessionHasErrors('photo');
```

6. Validation Test

Validasi harus gagal sebelum data masuk database.

```
public function test_report_cannot_be_created_with_empty_fields(): void
{
    $user = User::factory()->reporter()->create();
    $response = $this->actingAs($user)->post('/reports', []);
    $response->assertSessionHasErrors([
        'area_id', 'category_id', 'location_detail',
        'description', 'urgency', 'photo'
    ]);
}
```

Uji juga manipulasi input, misalnya urgency = super_urgent. Sistem harus menolak karena nilai tidak termasuk low, medium, high, atau emergency.

7. Role & Security Testing

Role Permission Matrix menjadi referensi untuk automated test maupun UAT.

Fitur	Reporter	Technician	Verifier	Admin
Dashboard	Ya	Ya	Ya	Ya

Fitur	Reporter	Technician	Verifier	Admin
Buat laporan	Ya	-	Ya	Ya
Lihat laporan sendiri	Ya	-	Ya	Ya
Task assigned	-	Ya	Ya	Ya
Verifikasi	-	-	Ya	Ya
Assign teknisi	-	-	Ya	Ya
Update pekerjaan	-	Ya	Ya	Ya
Master Area	-	-	-	Ya
Master Category	-	-	-	Ya
Master User	-	-	-	Ya

7.1 Reporter tidak boleh akses master data

```
$reporter = User::factory()->reporter()->create();
$this->actingAs($reporter)->get('/areas')->assertStatus(403);
```

7.2 Admin boleh akses

```
$admin = User::factory()->admin()->create();
$this->actingAs($admin)->get('/areas')->assertStatus(200);
```

Menyembunyikan menu Blade saja tidak cukup. Pembatasan wajib diterapkan juga pada route/middleware/policy.

8. Data Access & Authorization

Rekomendasi hak akses data: Reporter hanya laporan miliknya; Technician hanya task yang di-assign; Verifier dan Admin dapat melihat seluruh laporan.

8.1 Controller check sederhana

```
$user = auth()->user();

if ($user->role === 'reporter' && $report->user_id !== $user->id) abort(403);
if ($user->role === 'technician' && $report->assigned_to !== $user->id) abort(403);
```

8.2 Lebih rapi menggunakan ReportPolicy

```
php artisan make:policy ReportPolicy --model=Report

public function view(User $user, Report $report): bool
{
    if (in_array($user->role, ['admin', 'verifier'])) return true;
    if ($user->role === 'reporter') return $report->user_id === $user->id;
    if ($user->role === 'technician') return $report->assigned_to === $user->id;
    return false;
}
```

9. Workflow Transition Test

Status tidak boleh melompat sembarangan. Contoh reported langsung menjadi completed harus ditolak.

```
public function canTransitionTo(string $newStatus): bool
{
    $transitions = [
        'reported' => ['verified', 'rejected'],
        'verified' => ['in_progress', 'rejected'],
        'in_progress' => ['waiting_material', 'completed'],
        'waiting_material' => ['in_progress', 'completed'],
        'completed' => [],
        'rejected' => [],
    ];

    return in_array($newStatus, $transitions[$this->status] ?? []);
}

if (! $report->canTransitionTo($validated['status'])) {
```

```

    return back()->withErrors(['status'=>'Perubahan status tidak diperbolehkan.']);
}

```

10. Business Rule Improvement

10.1 Assignment hanya ke Technician

```

$technician = User::where('id',$validated['assigned_to'])
->where('role','technician')
->firstOrFail();

```

10.2 Completion photo wajib

```

if ($request->status === 'completed') {
    $request->validate([
        'completion_photo'=>'required|image|max:5120',
    ]);
}

```

10.3 Nomor laporan lebih mudah dibaca

Ubah format timestamp menjadi nomor berurutan per tahun, misalnya CF-2026-000001.

```

$year = now()->year;
$lastReport = Report::whereYear('created_at',$year)->latest('id')->first();
$sequence = $lastReport ? ((int) substr($lastReport->report_number,-6)) + 1 : 1;
$reportNumber = sprintf('CF-%s-%06d',$year,$sequence);

```

11. Mobile & User Experience Testing

Karena laporan sangat mungkin dibuat melalui smartphone, lakukan test responsive minimal pada ukuran berikut:

Perangkat	Resolusi acuan
Desktop	1920 x 1080
Laptop	1366 x 768
Tablet	768 x 1024
Mobile	390 x 844

- Sidebar dapat dibuka/tutup.
- Tombol Kirim mudah ditekan.
- Input foto mudah digunakan.
- Dropdown tidak keluar layar.
- Tabel tetap dapat discroll tanpa merusak layout.
- Gambar tidak terlalu besar.
- Status dan progress tracker mudah dibaca.

Untuk mobile, tombol submit dapat menggunakan class btn-block agar area klik lebih besar.

12. Form & Visual Improvement

12.1 Tampilkan validation error di field

```

class="form-control @error('area_id') is-invalid @enderror"

@error('area_id')
<div class="invalid-feedback">{{ $message }}</div>
@enderror

```

12.2 Pertahankan old input

```

<option value="{{ $area->id }}" {{ old('area_id') == $area->id ? 'selected' : '' }}>
    {{ $area->name }}
</option>

```

12.3 Human-readable status

```
public function getStatusLabelAttribute()
{
    return match ($this->status) {
        'reported'=>'Dilaporkan',
        'verified'=>'Diverifikasi',
        'in_progress'=>'Dalam Penanganan',
        'waiting_material'=>'Menunggu Material',
        'completed'=>'Selesai',
        'rejected'=>'Ditolak',
        default=>$this->status,
    };
}
```

13. Dashboard & SLA Testing

Verifikasi angka dashboard terhadap data database. Jangan hanya memeriksa tampilannya.

```
$this->assertDatabaseCount('reports',10);
```

Setelah core test lolos, dashboard dapat ditambah KPI: Emergency Open, Waiting Material, Average Resolution Time, Completion Rate, dan Overdue Report.

13.1 SLA berdasarkan urgency

Urgency	Target
Darurat	< 2 jam
Tinggi	< 24 jam
Sedang	2-3 hari
Rendah	Pemeliharaan rutin

Status SLA dapat dikembangkan menjadi ON TIME, NEAR SLA, dan OVERDUE.

14. Logging dan Penanganan Error

Jika terjadi HTTP 500, cek file berikut sebelum mengubah kode secara acak:

```
storage/logs/laravel.log
```

Catat error, route, user, langkah reproduksi, dan timestamp agar troubleshooting dapat diulang.

15. User Acceptance Test (UAT)

Minimal libatkan satu orang untuk setiap role: Reporter, Verifier/Sarpras, Technician, dan Admin.

15.1 UAT Reporter

- Login
- Buat laporan
- Upload foto
- Pilih area/kategori/urgency
- Submit
- Buka laporan
- Cek progress
- Logout

15.2 UAT Verifier

- Login

- Buka laporan baru
- Verifikasi
- Assign technician
- Lihat histori
- Monitoring laporan

15.3 UAT Technician

- Login
- Buka My Task
- Buka detail
- Update In Progress
- Waiting Material bila diperlukan
- Upload foto selesai
- Update Completed

15.4 UAT Admin

- Dashboard
- Master Area
- Master Category
- Manage User
- Semua laporan
- Monitoring status
- Review KPI dashboard

16. Format UAT

ID	Modul	Skenario	Expected Result	Status
UAT-001	Login	Login valid	Dashboard tampil	
UAT-002	Login	Password salah	Error tampil	
UAT-003	Report	Buat laporan	Data tersimpan	
UAT-004	Photo	Upload JPG	Foto tersimpan	
UAT-005	Role	Reporter akses Area	403	
UAT-006	Assign	Assign Technician	PIC tersimpan	
UAT-007	Status	Start work	In Progress	
UAT-008	Complete	Upload hasil	Completed	
UAT-009	Timeline	Buka report	History tampil	
UAT-010	Dashboard	Report selesai	KPI berubah	

17. Bug Tracker dan Improvement Register

17.1 Bug Tracker

Bug ID	Modul	Problem	Severity	Status
BUG-001	Login	Error message tidak tampil	Medium	Open
BUG-002	Report	Foto >5 MB masih masuk	High	Open
BUG-003	Role	Reporter dapat akses Area	Critical	Open
BUG-004	Mobile	Table overflow	Low	Open

Severity: Critical = keamanan/data loss/aplikasi tidak dapat digunakan; High = fitur utama gagal; Medium = fitur masih usable tetapi bermasalah; Low = cosmetic/UI.

17.2 Improvement Register

IMP ID	Improvement	Priority	Phase
IMP-001	Filter laporan	Medium	Tahap 3
IMP-002	Search report ID	High	Tahap 3
IMP-003	Mobile camera	High	Tahap 3
IMP-004	SLA warning	Medium	Next
IMP-005	Email notification	Low	Next

18. Improvement Prioritas Sebelum Finalization

- Search report number
- Filter status, area, dan urgency
- Pagination
- Preview foto sebelum submit
- Confirmation sebelum Complete
- Completion photo wajib
- Report Policy / authorization
- Validasi status transition
- Responsive mobile
- Status label yang mudah dibaca
- Report number yang lebih ringkas
- Flash notification success/error
- Dashboard realtime dan KPI dasar

19. Command Testing

```
php artisan test
php artisan test tests/Feature/ReportTest.php
php artisan test --filter=ReportTest
```

Target bukan sekadar 0 failed. Pastikan skenario kritis benar-benar memiliki test.

20. Minimum Test Case

No	Kelompok	Skenario
1-4	Authentication	Login valid, login invalid, logout, guest dashboard
5-13	Reporter	Create report, field wajib, upload, file size, urgency, own/other report

No	Kelompok	Skenario
14-18	Verifier	View all, verify, assign technician, invalid assignee, reject
19-24	Technician	Assigned task, access control, start, waiting, complete, photo wajib
25-27	Workflow	Invalid transition, valid transition, history recorded
28-30	Admin	Master data, dashboard counts, role access

21. Checklist Tahap 3

No	Testing / Improvement	Status
1	Database testing terpisah	☐
2	php artisan test berjalan	☐
3	User Factory	☐
4	Area Factory	☐
5	Category Factory	☐
6	Report Factory	☐
7	Login Test	☐
8	Logout Test	☐
9	Guest Access Test	☐
10	Create Report Test	☐
11	Photo Upload Test	☐
12	Validation Test	☐
13	File Size Test	☐
14	Role Permission Test	☐
15	Reporter Data Access Test	☐
16	Technician Access Test	☐
17	Assignment Test	☐
18	Workflow Transition Test	☐
19	History Test	☐
20	Completion Photo Test	☐
21	Dashboard Test	☐
22	Search & Filter	☐
23	Mobile UI Test	☐
24	UAT Reporter	☐
25	UAT Verifier	☐
26	UAT Technician	☐
27	UAT Admin	☐
28	Bug Register	☐
29	Improvement Register	☐
30	Regression Test	☐

22. Milestone Tahap 3

Milestone	Isi
3.1 Automated Basic Test	Login + database + create report + validation + upload
3.2 Security & Workflow	Role + authorization + assignment + transition + history
3.3 User Experience	Responsive + search + filter + messages + status label
3.4 UAT	Reporter + Verifier + Technician + Admin
3.5 Bug Fix & Regression	Fix + retest + close bug

23. Definition of Done

- Semua fitur utama berjalan.
- Tidak ada Critical Bug.
- Tidak ada High Bug terbuka.
- Role access dan data authorization benar.
- Input invalid tidak masuk database.
- Upload file tervalidasi.
- Status tidak dapat dilompati sembarangan.
- Completion memiliki bukti foto.
- History lengkap.
- Dashboard sesuai database.
- Tampilan usable di smartphone.
- UAT empat role selesai.
- Bug yang diperbaiki sudah regression test.